ALICE L WALTON SCHOOL OF MEDICINE

α1–Adrenergic Blockers

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Adapted from: α-Adrenergic Agents

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Classify alpha blockers into uroselective and non-uroselective
2. Explain the mechanism of action, and clinical uses of uroselective and non-uroselective alpha blockers.
3. Explain the adverse effects associated with the use of uroselective vs. non-uroselective alpha blockers.
4. Explain the mechanism of action, and clinical uses of phenoxybenzamine and phentolamine.

CASE CONNECTION

CW is a 64-year-old male with benign prostatic hypertrophy who is taking tamsulosin for his urinary symptoms and reporting great improvement. He recently began a new relationship and seeks your advice about erectile dysfunction, saying "I wonder if my prostate medicine is making that worse." He inquires about stopping that or starting another medicine for his erectile dysfunction.

How will you counsel CW about the sexual and urinary effects of his tamsulosin and potentially adding a phosphodiesterase inhibitor? Consider your answers as you read, and we'll revisit CW at the end of the brick.

GO TO CONCLUSION ↓

α1–RECEPTORS

α1-Receptors are found in the smooth muscle arterioles, the gastrointestinal and urinary sphincters, the urethra, the prostate gland, and the dilator pupillae (radial) muscle of the eye. α1-Receptors are divided into three subtypes: α_{1A}, α_{1B}, α_{1D}. It is very important to note that α_{1A} is the subtype that is predominately expressed in the prostate smooth muscle. All α1-Receptors belong to the G_q type of GPCR. Hence, most of the actions are mediated by Ca^{2+} and CaM-sensitive pathways. For review of GPCRs refer to the Autonomic Nervous System- Vas

WHAT ARE α1–BLOCKERS?

The sympatholytic drugs that affect α1-receptors are called α1-antagonits or α1-blockers

ACTIVE LEARNING QUESTION

The inhibition of Ca and CaM-sensitive pathways in the vascular smooth muscle can lead to

A. ○ vasoconstriction

B. ● vasodilation

Explanation

Vasodilation (B). An increase in cytosolic Ca increases CaM activation which subsequently increases actin-myosin binding leading to smooth muscle contraction of the vasculature. Inhibition of this pathway can reverse the muscle contraction, causing vasodilation.

HOW DO YOU CLASSIFY A1–RECEPTOR BLOCKERS? (LO–1)

The α1-blockers fall into two main clinical categories based on receptor selectivity: α1-selective and nonselective (which act on both α1-receptors and α2-receptors). Common agents are summarized in Table 1. (modify brick table). All α-1 blocker drugs end with the suffix "-osin". α1-Receptor blockers can also be classified based on their clinical use as uroselective and non-uroselective. Note that the table includes non-selective α-blockers, phenoxybenzamine and phentolamine. Table 1.

CLASSIFICATION	DRUGS
RECEPTOR SELECTIVITY	
Nonselective α-blockers	Phenoxybenzamine, Phentolamine
Nonselective α1-blockers	Prazosin, Terazosin, Doxazosin, Alfuzosin
α_{1A} receptor blocker	Tamsulosin
CLINICAL USE	
Uroselective α1-blockers	Tamsulosin, Alfuzosin
Non-uroselective α1-blockers	Prazosin, Terazosin, Doxazosin

Table 1: Classification of α-blockers by selectivity and use

Which are the Uroselective α1-blockers?

HOW DO THESE α1–BLOCKERS WORK? (LO–2)

As stated earlier, α1-blockers elicit their actions by inhibiting the function of α1-receptors. The mechanism of action of these agents is easier to understand within the context of their clinical indications. It also provides the rationale for their indications and use. α1-blockers were primarily approved for the treatment of hypertension and benign prostatic hyperplasia. Non-selective α-blocker are used in pheochromocytoma.

Benign Prostatic Hyperplasia

α1-blockers (eg, tamsulosin, alfuzosin, silodosin) are used to treat urinary symptoms associated with benign prostatic hyperplasia (BPH), specifically urinary hesitancy, decreased stream, and incomplete voiding. Activation of α1-receptors in prostatic smooth muscle, the prostatic urethra, and the bladder neck means that stimulation causes urethral and sphincter smooth muscle contraction, which de tatic portion of the urethra. In BPH, cellu obstruction of this same portion of the urethra, resulting in difficulty urinating. Blocking these α1-receptors can prevent smooth muscle contraction, dilate the urethra, and decrease the resistance to urine flow (Figure 1). α_{1A} receptors expression in the prostate and urinary tract is considerably high; thus, Tamsulosin, an α1A blocker (also Uroselective), is often prescribed in the treatment of BPH. Other uroselective α1-blockers such as Alfuzosin are also used in the BPH treatment.

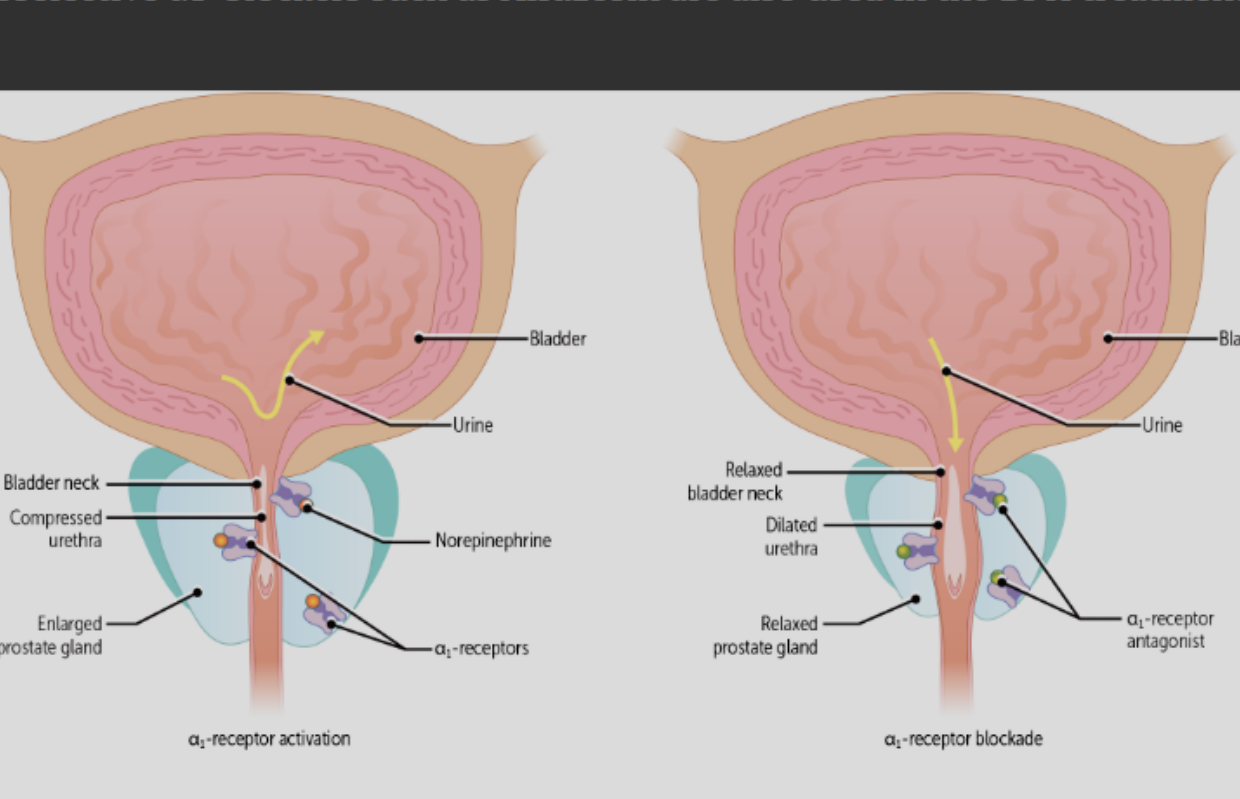

Figure 1 By selectively blocking these α_1-receptors with drugs such as tamsulosin, we can prevent smooth muscle contraction, dilate the urethra, and decrease the resistance to urine flow

How do α_1-receptor blockers reduce the urinary symptoms of benign prostatic hyperplasia?

Hypertension

α1-receptor blockers (e.g., doxazosin) inhibit the activation of postsynaptic α1-Receptors. By inhibiting α1-Receptor function, peripheral vascular resistance is decreased. Thus, blood pressure decreases. α1-blockers may be used to lower blood pressure in patients with hypertension. However, they are not first-line agents because of their unfavorable safety profile compared with other antihypertensive agents. Non-uroselective α1-blockers, Prazosin, Doxazosin, and Terazosin are used as anti-hypertensive agents (again not used as first line!!). Doxazosin and Terazosin also have been approved for the treatment of BPH.

Pheochromocytoma (LO–4)

Pheochromocytoma is a very rare neuroendocrine, catecholamine-producing tumor usually found in the adrenal medulla. A flood of endogenous catecholamines, whic rs, can result in a hypertensive crisis associated with flushing and tachycardia, although many patients just have high blood pressure. The drugs phenoxybenzamine and phentolamine are the most common agents used for the initial management of a pheochromocytoma before surgical resection. They are both nonselective antagonists of the α-receptors. They differ in that phenoxybenzamine is an irreversible antagonist, resulting in much longer and more pronounced effects, while phentolamine is reversible. By blocking both α1-receptors and α2-receptors, these drugs can decrease total peripheral resistance and blood pressure. Phenoxybenzamine is used for long-term control in patients who are inoperable, where phentolamine is used in patients for short control of hypertension in pheochromocytoma patients.

CLINICAL CORRELATION

The medical management of pheochromocytoma is initiated with α_1-blockade and followed by the addition of β-blockers. This order of drug administration is critical because β- blockade will knock out the vasodilatory effects of β-adrenergic receptors, leading to unopposed α-mediated vasoconstriction and a hypertensive crisis. This drug interaction is often seen in older male patients with BPH (benign prostatic hyperplasia). Many of these patients will take tamsulosin, an α_1-blocker, but they may also struggle with erectile dysfunction, due to age, comorbid conditions, etc. A physician may be tempted to prescribe sildenafil (a PDE-5 inhibitor that we will discuss later), which could cause dangerous hypotension when combined with tamsulosin.

WHAT ADVERS H α–BLOCKERS? (LO–3)

- ***Orthostatic Hypotension (postural hypotension)*** Orthostatic hypotension (OH) is one of the main adverse effects of this drug class. OH is most likely caused by the inhibition of vascular smooth muscle contraction, which subsequently leads to vasodilation. The compensatory baroreflex fails to increase sympathetic outflow, thus causing syncope and dizziness. OH can lead to an increased risk for falls, especially in frail older adult populations.
    - All α1-blockers (uro- or non-uroselective) carry the risk of OH.
    - Risks of OH include first dose or redose after D/C, rapid increase in dose, concurrent medications that can cause OH (nitrates, PDE-5 inhibitors), and type of α1-blocker. Tamsulosin has the lowest risk due to its selective affinity for α_{1A} receptors. Prazosin has a higher affinity for α1-receptors and a rapid onset of action, thus increasing the risk of OH. The risk of OH is higher within the first 24 hours of the first dose. Patients should be instructed to rise slowly from sitting or supine positions as long as they are on these medications.
- ***Intraoperative Floppy Iris Syndrome (IFIS)*** IFIS is caused by the inhibition of iris dilator smooth muscle contraction caused by α1-receptor inhibition and atrophy. It is most often seen in patients undergoing cataract surgery. Tamsulosin is particularly associated with IFIS (carries highest risk), along with other α1-blockers.
- ***Other ADRs*** Some α1-blockers (e.g., tamsulosin, silodosin) relax the bladder and can cause ejaculation difficulties. Without the ability to contract smooth muscle cells and propel semen forward, retrograde ejaculation can result in sperm being propelled backwards into the bladder. Priapism and pr ted (associated with non-selective α1-blockers).

All α1-blockers increase the risk for fall in elderly because they can cause __________.

CLINICAL CORRELATION

The combination of α_1-receptor blockers with other vasodilatory agents, like blood pressure medications, may cause significant hypotension. Concomitant use of α-blockers with phosphodiesterase-5 inhibitors (eg, sildenafil) can precipitate severe hypotension, which may lead to loss of consciousness.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to CW, and urinary effects of his tamsulosin and potentially adding a phosphodiesterase inhibitor?

You say that although there are sometimes sexual side effects (such as retrograde ejaculation) with tamsulosin, it doesn't generally affect erections (because it is a sympathetic drug and erections are primarily mediated by the parasympathetic system). So stopping it won't likely help his erections. However, because phosphodiesterase inhibitors also cause vasodilation, the combination of these drugs with tamsulosin can be dangerous, leading to hypotension. CW thanks you for the information and opts to discuss this with his partner before deciding what to do next. You assure him that you will work with him to find the best solution.

Summary

- α1-Receptors are found in smooth muscle arterioles, gastrointestinal and urinary sphincters, the urethra, the prostate gland, and the dilator pupillae muscle of the eye.
- α1-blockers are sympatholytic drugs that can be classified based on receptor selectivity and clinical use. They are primarily used for the treatment of hypertension, benign prostatic hyperplasia, and pheochromocytoma.
- α1-blockers work by preventing smooth muscle contraction, dilating the urethra, and decreasing the resistance to urine flow. They also decrease peripheral vascular resistance, thereby reducing blood pressure.
- In the treatment of pheochromocytoma, α1-blockers decrease total peripheral resistance and blood pressure. They are used for initial management before surgical resection.
- Common adverse ss, orthostatic hypotension, drowsiness, weakness, and headache. They can also cause ejaculation difficulties.
- α2-Agonists block norepinephrine release in the CNS and can be used to treat hypertension, attention-deficit/hyperactivity disorder, and substance withdrawal. Their main adverse effect is sedation.

Review Questions

1. Which of the following structures is most affected by α_1-blockade?

A. ○ Adrenal gland

B. ○ Cerebral cortex

C. ○ Liver

D. ● Prostate

Hide Explanation

The correct answer is the prostate (D). α_1-Blockade can target the receptors on the prostate, urethra, and bladder neck to dilate the urethra (e.g., in benign prostatic hyperplasia). The adrenal gland has no notable α_1-receptors (A). Although there are α-receptors in the cerebral cortex, this is not clinically relevant (B). α-Blockade does not occur in the liver (C).

2. Which α1-blocker is uroselective?

A. ● Tamsulosin

B. ○ Prazosin

C. ○ Terazosin

D. ○ Doxazosin

Hide Explanation

The correct answer is Tamsulosin (A). Tamsulosin is an α1A blocker that is often prescribed for BPH due to its high expression in the prostate and urinary tract. Alfuzosin (B) is also used in BPH treatment but is not as selective. Prazosin (C) and Terazosin (D) are non-selective α1-blockers. Doxazosin (E) is another non-selective α1-blocker.

3. From the list below, which drug has the lowest risk of orthostatic hypotension (OH)?

A. ● Tamsulosin

B. ○ Terazosin

C. ○ Prazosin

D. ○ Doxazosi

Hide Explanation

The correct answer is tamsulosin (A). Tamsulosin has the lowest risk of causing OH because it is has α1A receptor selectivity. Terazosin (B), Prazosin (C), and Doxazosin (D) α1- non-selective blockers that can decrease BP by inhibition of peripheral smooth muscle contraction leading to OH.

4. Phenoxybenzamine is used in treating Pheochromocytoma because it is a

A. ○ Non-selective α1-blocker

B. ● Non-selective α-blocker

C. ○ Selective α1-blocker

D. ○ Selective α2-blockers

Hide Explanation

The correct answer is Non-selective α-blocker (B). Because of its non-selective properties, Phenoxybenzamine can inhibit the sympathetic tone and decrease the catecholamine surge caused by the adrenal tumor. The selective inhibition of α1 (C) or α2 (D) does not decrease catecholamine release and hence is not effective. Phenoxybenzamine does not selectively inhibit α1 receptor function.

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Classify alpha blockers into uroselective and non-uroselective~~
02 ☑ ~~Explain the mechanism of action, and clinical uses of uroselective and non-uroselective alpha blockers.~~
03 ☑ ~~Explain the adverse effects associated with the use of uroselective vs. non-uroselective alpha blockers.~~
04 ☑ ~~Explain the mechanism of action, and clinical uses of phenoxybenzamine and phentolamine.~~

Objective Confidence: 4/4